

Plant Leaf Disease Detection Using Hybrid Deep Learning Framework

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Abstract—Plant diseases are a problem in farming today. They affect how much crops grow, their quality and the food we have. If we do not find infections early they can cause a lot of money loss and less crops. Usually finding plant diseases is done by experts looking at them manually. This way is slow. Can be different from person, to person. These problems show we need a system that can automatically detect diseases accurately and fast. Plant diseases are an issue and we need to find them quickly to save crops.

In This paper is about a system that uses deep learning to figure out what is wrong with plant leaves just by looking at pictures. The system uses two things: EfficientNetV2-S, which's really good at looking at pictures quickly and MobileViTv2 which is a simpler way of looking at the whole picture to see how everything is connected. The system also has a way of combining the information it gets from these two things so it can decide what is most important and use that to make a decision. The plant leaf diseases are figured out by the system using this combination of information from the pictures. The system we are suggesting is. Tested with a special set of pictures of Apple and Grape leaves.

The model is fine-tuned with a tool called the Adam optimizer. We check how well the model is doing by looking at things like accuracy, precision, recall and F1-score. The model is trained to recognize Apple and Grape leaves and tell if they are healthy or diseased. We use these metrics to see how good the model is, at identifying diseased Apple and Grape leaves. Experimental results show that our hybrid model does really well, with an accuracy of up to 96.67%. It keeps performing across several tests. We also looked into how our model works using Grad-CAM visualization. It focuses on the areas that are affected by disease. This helps us trust the models predictions.

Overall, the proposed method is a solution that combines accuracy, speed and understandability. This method works with different disease patterns making it suitable for real-world use in smart farming systems. The framework helps farmers and specialists find diseases early allowing for action and leading to improved crop management and sustainability. Farmers and specialists can discover diseases early using this framework, which enables intervention and contributes to better crop management and sustainability. The proposed method provides a solution that combines precision, efficiency and interpretability making it suitable for use, in smart agriculture systems.

Index Terms—Plant Leaf Disease Detection, Deep Learning, Convolutional Neural Network (CNN), Transformer, EfficientNetV2-S, MobileViTv2, Hybrid Model, Feature Fusion, Image Classification, Grad-CAM

I. INTRODUCTION

Agriculture is very important to the world because it helps feed the people. The problem is that plant diseases are an issue. They can hurt the quality and amount of crops that farmers grow. These diseases can spread fast when the weather is just right like when it's humid or when the temperature changes a lot. So it is very important to catch these diseases. If farmers do not see the problem away it can cause a lot of damage to their crops and they can lose a lot of money. Agriculture is critical because it helps feed the worlds rising population.

Traditionally, Agricultural professionals usually do things the way and check plants by hand to find diseases. This way works well but it takes a lot of time and people to do it. The person who is checking also needs to have a lot of experience. When you have a farm it is hard to keep an eye on everything all the time. This means you might find out about a disease late or get it wrong especially when different diseases look similar. Agricultural professionals need a way to find diseases that is faster and more accurate. Agricultural professionals need automated solutions to help them find diseases, in plants. This will help agricultural professionals detect plant diseases.

With advancements in Artificial Intelligence and Deep Learning detecting plant diseases using images has become a good solution. Deep learning models, Convolutional Neural Networks are good at classifying images. These models can. Find important details in images like color changes, texture patterns and structural changes in plant leaves. However Convolutional Neural Networks focus on parts of an image and might miss the bigger picture. This can make it hard for them to recognize patterns of disease. Artificial Intelligence and Deep Learning are really helping us with this. Convolutional Neural Networks are a part of this. They help us classify images of plant leaves. Plant diseases can be detected using these images.

To fix this problem new models have been created. These models are based on transformer architecture. They can understand images better by looking at all parts of the image. These models use a technique called attention. It helps them see how different parts of an image are connected. By

combining models like CNNs with these new transformer models we get the best of both worlds. The old models are good at finding details in images. The new models are good, at understanding the picture. This combination makes the models work better. We use CNNs for feature extraction. The transformer models help with understanding of the image context. This results in performance.

II. LITERATURE REVIEW

Artificial Intelligence and Machine Learning are being used to analyze images. Convolutional Neural Networks are often used for this. They can pick up on details in images all by themselves. Studies have found that Convolutional Neural Networks do a job than older methods, like SVM, KNN and Random Forest when it comes to accuracy. Artificial Intelligence and Machine Learning are helping to improve the identification of plant diseases [5]. They also perform better when trained on large image datasets [11]. Further improvements in CNN architectures have enhanced their ability to detect plant diseases more effectively [8]. Hybrid AI and ML approaches have also been successful in identifying diseases and assessing their severity [10].

Several studies have demonstrated the effectiveness of CNN models in real-time plant disease detection. Improved CNN architectures have been successfully applied to detect apple leaf diseases with high accuracy [3]. Similarly, CNN-based approaches have shown strong performance under varying environmental conditions [7]. In grape leaf disease detection, deep CNN models have effectively identified multiple disease classes even with complex backgrounds [12]. Domain-specific models such as MaizeNet further improve performance by adapting architectures for specific crops [6].

Researchers have focused on improving model efficiency and accuracy using advanced architectures such as EfficientNet, which balances network depth, width, and image resolution [9]. Hybrid approaches that combine EfficientNet with other models have shown improved performance [1]. Comparative studies indicate that such hybrid models are highly effective for classification tasks. Recent work also combines detection frameworks with classification models to enhance real-world applicability.

Another important research area is multi-scale feature learning. Feature Pyramid Networks (FPN) extract features at multiple resolutions, enabling models to capture both fine details and global patterns. These models have shown strong performance in crop disease detection. Advanced architectures like MSP-Net improve performance in complex environments, while MCPDNet enhances detection by combining multi-scale features. However, these models often require careful optimization due to their complexity. Attention mechanisms and feature fusion techniques have further improved performance. Lightweight residual networks with attention modules help focus on important disease regions [4]. Multi-scale attention-based models have shown strong results in crops like tomato. Feature fusion methods improve both interpretability and accuracy. These techniques have also been

applied successfully in rice disease detection. Other studies explore hierarchical and pyramid-based fusion methods, as well as self-attention-based approaches for capturing long-range dependencies. Multi-scale feature fusion continues to produce strong results in disease detection tasks.

Despite these advancements, challenges still remain. Many deep learning models require high computational resources and are sensitive to parameter tuning. Detection models such as YOLO also face challenges in balancing accuracy and speed in real-time applications.

In this project, two approaches are explored. The first is a Multi-Scale Feature Pyramid Network (FPN) with dynamic scale weighting. The second is a hybrid model combining EfficientNetV2-S and MobileViTv2 using gated feature fusion.

III. RESEARCH GAP

The use of learning for plant disease detection has come a long way.. There are still some big problems with how it is being used. Most of the time people who study this are trying to make the detection more accurate. They are not thinking about important things like making sure the model works well every time it does not use too much computer power and it can be used in the real world. This means that what works in a study does not always work on a farm. Plant disease detection with learning is still not very good at being used in the real world where it is really needed.

Multi-scale architectures such as Feature Pyramid Networks (FPN) have been widely used to enhance feature extraction by capturing information at different resolutions. While these models improve representation capability, they introduce increased architectural complexity, higher memory consumption, and longer training time.

A big problem is that we do not have a lot of kinds of agricultural data to work with. When we use deep learning models that have a lot of parameters and we train them on an amount of data they do not work well with new data they have not seen before. Deep learning models and agricultural datasets are important here. The methods we use to make these models better like feature fusion and attention mechanisms can be helpful but they also make the model use more computer power and we have to be careful when we set up the model or it will not work as well as it should.

Moreover many existing methods do not fully use the benefits of pretrained models. This results in training times and more data needs. There is also not balance between accuracy and efficiency. Accurate models are often very expensive to run and hard to use in places with limited resources, like farms and rural areas.

IV. METHODOLOGY

The suggested method uses a hybrid deep learning approach to automatically detect the plant leaf diseases. It scans photos of plant leaves and divides them into categories such as healthy or damaged. The system combines fast

feature extraction with attention-based learning to improve classification accuracy, stability, and generalizability.

The procedure has several stages, including dataset collection, preprocessing, feature extraction, model training, and prediction. This study considers two models: a Multi-Scale Feature Pyramid Network (FPN) and a hybrid model that combines EfficientNetV2-S and MobileViTv2. Both models are tested under the same conditions to determine the best effective method for detecting plant diseases.

Although the Feature Pyramid Network or FPN is good at capturing features at levels it has some practical limitations. The model gets more complicated because it uses feature scales, which makes it cost more to compute and takes longer to train. I noticed during testing that the FPN model was unstable during training with swings in how accurate it was on validation data. This made it hard to get results. The way the model handles scales also makes it picky about the settings used which makes it harder to optimize. Another issue is that the model might combine features in a way that's not helpful which can hurt how well it generalizes, especially with smaller datasets.

Considering these challenges, the EfficientNetV2-S and MobileViTv2 hybrid model was selected as the primary approach. This model provides a better balance between performance and efficiency by combining strong local feature extraction from EfficientNetV2-S with global contextual understanding from MobileViTv2.

A. System Overview

The proposed system uses a step-by-step process to detect plant diseases. The process starts with collecting a dataset and organizing it into folders based on disease classes. The input images are then prepared to have the size and quality. This preparation step ensures that all images are consistent. The prepared images are then fed into deep learning models to extract features and classify diseases. The system uses two models: EfficientNetV2-S and MobileViTv2.

EfficientNetV2-S finds features like texture and color changes in the images. MobileViTv2 looks at the picture and finds relationships using attention mechanisms. The features from both models are combined to get a picture of the input image. This complete picture helps the system to make an accurate prediction. The system then outputs the predicted disease class and a confidence score.

The system also uses visualization tools like Grad-CAM. Grad-CAM highlights the parts of the image that affect the prediction. This helps to make the system more understandable and reliable. The plant disease detection system is more accurate and reliable, with these features. The system helps to detect plant diseases effectively.

B. Dataset Description

The PlantVillage dataset is what we used for this study. This dataset is really popular when it comes to finding diseases in plants. It has a lot of pictures of plant leaves that

are labeled and these pictures show leaves that're healthy and leaves that are not.

We chose to look at pictures of Apple leaves and Grape leaves for this project. This way we can really focus on these two types of leaves. See how different models work. The PlantVillage dataset has pictures of leaves with different types of diseases like Apple Scab and Grape Black Rot. We also have pictures of leaves.

The PlantVillage dataset does not have the number of pictures for each type of disease. This is, like what happens in the world. To deal with this we use techniques when we are training our model. These techniques help our model learn from the pictures we have so it can get better at finding diseases in leaves.

Class	Image Count
Apple Healthy	1645
Apple Scab	587
Apple Black Rot	621
Apple Cedar Apple Rust	275

Fig. 1. Apple Dataset Distribution

Class	Image Count
Grape Black Rot	1180
Grape Esca (Black Measles)	1383
Grape Leaf Blight (Isariopsis Leaf Spot)	1076
Grape Healthy	423

Fig. 2. Grape Dataset Distribution

1) *Data Preprocessing*: Data preprocessing is an important step. It helps make the input data better more consistent and reliable. This is done before the data is used to train a deep learning model. All images are made the size 128×128 pixels. This makes it easier to process them. Keeps the important details like texture, color and shape. This way all the inputs are the same, which helps when processing images at once during training.

The pixel values are to be adjusted to be between 0 and 1. This helps the training process go smoothly and finish faster. By making all the input values similar the model can learn better. It is not affected by differences in pixel brightness. To make the model even better we use data augmentation. This involves changing the images in ways.

We rotate them Flip them horizontally and vertically Shift them Zoom in and out and Make distortions These changes make the dataset bigger and more varied. The model can learn from different versions of the same image. Data augmentation is very helpful. It makes the model more robust. It helps

the model adjust to real-world changes like Different leaf orientations in Lighting conditions Scale and Background noise This reduces overfitting. It helps the model work well with unseen data. Preprocessing also helps remove noise and irrelevant changes, in the dataset.

2) *model Architecture*: The system they are talking about uses a kind of deep learning architecture. This architecture combines the things about convolutional neural networks and transformer-based models. The goal is to get the most out of plant leaf images. They want to look at both the details and the big picture to make the system better at telling plants apart.

The first part of the model is called EfficientNetV2-S. This part is really good at finding the things in images like edges and colors. It also finds the not simple things like textures. The reason they chose EfficientNetV2-S is that it can do a job with fewer instructions. This makes it faster for the computer to use. The second part is called MobileViTv2. This is a transformer-based model that looks at the whole image. It tries to understand how all the parts of the image work together. This is different, from neural networks which just look at the small parts. MobileViTv2 looks at the thing and tries to learn from it by paying attention to the important parts.

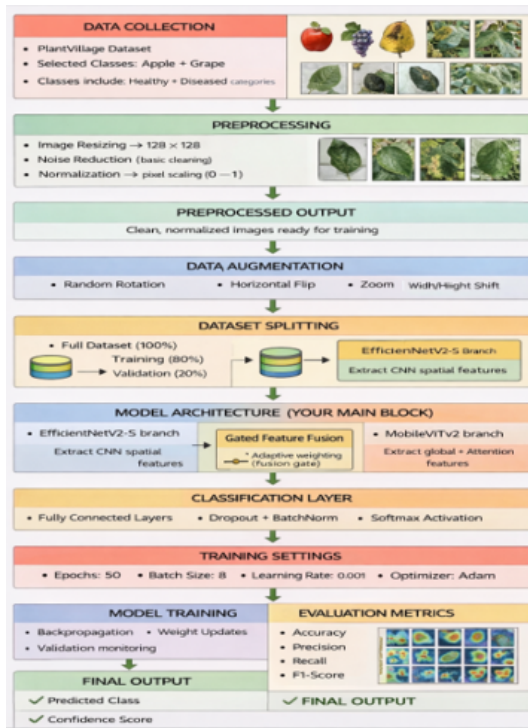


Fig. 3. Model Architecture

C. Hybrid Model Architecture

The model is designed to be good at looking at pictures and figuring out what is going on. It uses EfficientNetV2-S and the attention mechanism to do this. EfficientNetV2-S is good at finding the things and the attention mechanism is good at

understanding the big picture. This makes the model good, at telling diseases apart.

EfficientNetV2-S Backbone: The model they are talking about is a mix of things. It uses something called EfficientNetV2-S. Adds an attention mechanism that is similar to MobileViTv2. EfficientNetV2-S is the main part that looks at the pictures and finds things like edges and colors. It uses -trained weights so it can learn faster and not take as long to train.

Attention-Based Module: A Multi-Head Self-Attention mechanism is used to capture global relationships between different regions of the image. This allows the model to understand contextual information beyond local patterns.

Feature Fusion Strategy: Features extracted from both components are combined into a unified representation. This helps the model learn both detailed and contextual information, which is essential for distinguishing similar disease classes.

Classification Layers: The fused features are passed through dense layers with batch normalization and dropout. Batch normalization stabilizes training, while dropout prevents overfitting. The final layer uses Softmax activation to generate class probabilities.

D. Model Training

The model is trained over multiple epochs using mini-batch gradient descent. During training, both training and validation performance are monitored.

To improve efficiency and prevent overfitting, several techniques are used: • **Early Stopping**: Stops training when validation loss stops improving • **Reduce Learning Rate**: Adjusts learning rate when performance stagnates • **Model Checkpoint**: Saves the best model based on validation accuracy These strategies ensure proper convergence and avoid unnecessary training.

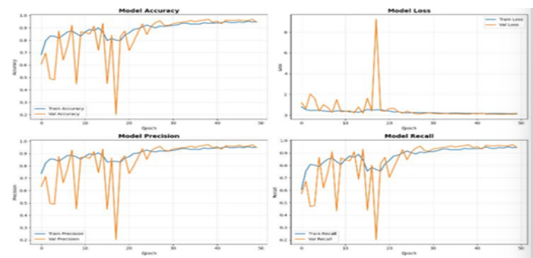


Fig. 4. Apple Dataset Training Graph

E. Model Evaluation

The trained model is tested with some validation data to see how well it works. The predictions that the model makes are compared to the labels to figure out how good the model really is.

F. Testing Phase

The testing phase is very important because it will show us how well the model can do its job of classifying things using data it has never seen before. The model learns from the data

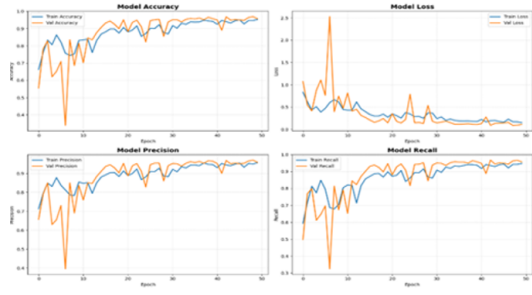


Fig. 5. Grape Dataset Training Graph

Metric	Apple Dataset	Grape Dataset
Training Accuracy	97.17%	96.68%
Validation Accuracy	96.64%	96.18%
Precision	96.63%	96.42%
Recall	96.48%	96.18%

Fig. 6. Performance Metrics of Hybrid Model

it is given during the training phase. The testing phase shows us how well the model can handle new and unknown things. For this study we used pictures of Apple and Grape leaves to test the model.

We use a different ways to measure how well the model is doing. Accuracy is the percentage of times the model gets it right. Precision is the number of guesses divided by the total number of guesses the model made. Recall is how good the model is at finding all the things it is supposed to find.

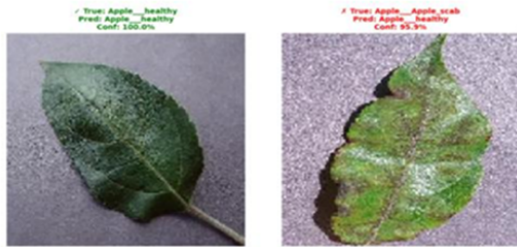


Fig. 7. Apple Image Prediction



Fig. 8. Grape Image Prediction

V. RESULTS

A. Performance Metrics

The performance of the proposed model is evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of the model's effectiveness in classifying plant leaf diseases across different categories.

Accuracy measures the proportion of correctly classified samples among the total number of predictions, providing an overall indication of model performance. Precision represents the correctness of positive predictions, ensuring that the model does not incorrectly label healthy leaves as diseased. Recall measures the model's ability to identify all relevant instances of a particular class, which is important for detecting all disease cases. The F1-score combines precision and recall into a single metric, offering a balanced evaluation of the model's classification capability.

Class	Precision	Recall	F1-Score
Apple Scab	0.9052	0.8974	0.9013
Apple Black Rot	1.0000	0.9839	0.9919
Apple Cedar Rust	0.9016	1.0000	0.9483
Apple Healthy	0.9785	0.9696	0.9740
Accuracy	96.16%		

Fig. 9. Classification Report for Apple Dataset

Class	Precision	Recall	F1-Score
Black Rot	0.9339	0.9576	0.9456
Esca (Black Measles)	0.9674	0.9674	0.9674
Leaf Blight	0.9952	0.9628	0.9787
Healthy	0.9882	1.0000	0.9941
Accuracy	96.67%		

Fig. 10. Classification Report for Grape Dataset

B. Grad-CAM

The Grad-CAM visualization is really useful because it shows us how the model makes predictions. It does this by highlighting the important parts of the pictures we give it. The model creates maps that show which parts of the picture are most important for its decisions. These maps clearly show that the model is paying attention to the parts of the leaves that're diseased like the dark spots and damaged areas. This means the model is actually learning about the diseases that affect plants than just looking at the background of the picture.

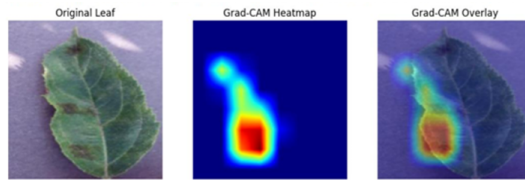


Fig. 11. Apple Leaf Grad-CAM

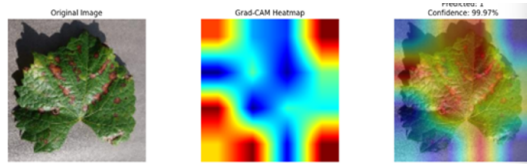


Fig. 12. Grape Leaf Grad-CAM

VI. CONCLUSION

This work presents a hybrid deep learning-based system for automatic plant leaf disease detection using image data. The proposed approach combines EfficientNetV2-S and MobileViTv2 to leverage both local feature extraction and global contextual learning. The system is designed to classify diseases in Apple and Grape leaves with high accuracy and reliability.

The model does well on many tests getting more than 96% accuracy on both sets of data. Training goes smoothly. The model works well even with data it has not seen before. This shows that it is quite robust. The Grad-CAM images also show that the model looks at the parts of the images that are affected by disease. This makes it easier to understand and trust its predictions. The model focuses on disease-affected regions, which improves interpretability and trust, in disease-affected regions.

The Multi-Scale Feature Pyramid Network model is not as good as the hybrid model we are talking about. The hybrid model is more stable it works better. It is more efficient. The Multi-Scale Feature Pyramid Network model is too complicated. It does not work well when we try to train it. This makes it not very useful for life situations. On the hand the hybrid model is a good balance of being accurate not using too much computer power and working well in general.

The system we are proposing is a way to detect plant diseases and it can be used in the real world to help farmers. It can help farmers find out if their plants are sick on and take better care of their crops.

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